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yeshwantsingh / swaragram

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swaragram / core.md

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Documentation of 'core' module

NAME swaragram.core

FUNCTIONS

```
computeSwaragram(stft, sampleRate=22050, winLength=2048, refPitch=69, refFreq=440)
    Computes a swaragram

    Args:
        stft (ndarray): Magnitude or power spectrogram
        sampleRate (int): Sampling rate
        winLength (int): Window size of Fourier fransform
        refPitch (int): Reference pitch (default: 69)
        refFreq (float): Frequency of reference pitch (default: 440.0)

    Returns:
        C (ndarray): Swaragram matrix

logFreqSpec(Y, sampleRate, winLength, refPitch, refFreq)
    Computes a log-frequency spectrogram

    Args:
        Y (ndarray): Magnitude or power spectrogram
        sampleRate (int): Sampling rate
        winLength (int): Window size of Fourier fransform
        refPitch (int): Reference pitch (default: 69)
        refFreq (float): Frequency of reference pitch (default: 440.0)

    Returns:
        Y_LF (ndarray): Log-frequency spectrogram

plotSwaragram(swara, t, save=False)
    Plot swaragram

    Args:
        swara: Swaragram matrix
        t: time indices
        save: save plot or not

poolMask(pitch, sampleRate, winLength, refPitch=69, refFreq=440)
    Return a mask for aggregating frequecy values

    Args:
        pitch (int): MIDI pitch value
        sampleRate (int): Sampling rate
        winLength (int): Window size of Fourier transform
        refPitch (int): Reference pitch (Tonic)(default: 69)
        refFreq (float):  Frequency of reference pitch (Tonic)(default: 440.0)

    Returns:
        im (ndarray): Set of frequency indices

swaraRange(pitch, bandwidth=0, refPitch=69, refFreq=440)
    Return range for a pitch value based on Shruti


    Args:
        pitch (int): MIDI pitch value
        bandwidth (int): Tolerance for the upper and lower frequency values of a given pitch
        refPitch (int): Reference pitch (Tonic) (default: 69)
        refFreq (float): Frequency of reference pitch (Tonic) (default: 440.0)

    Returns:
        lower (float): Lower value of the pitch frequency adjusted with bandwidth
        upper (float): Upper value of the pitch frequency adjusted with bandwidth
```

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swaragram / utils.md

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Documentation of 'utils' module

NAME swaragram.utils

FUNCTIONS

```
computeRefPitch(tonic)
    Compute tonic pitch and its frequency

    Args:
        tonic (float): Tonic of the audio sample

    Returns:
        tonicMidiPitch (int): Nearest midi pitch to tonic
        midiPitchFreq (float): Corresponding frequency to nearest midi pitch

computeSTFT(x, win_length=2048, hop_length=512, window='hann', pad_mode='constant')
    Calculate STFT of an audio signal

    Args:
        x (np.array): Audio sample values sampled at sampleRate
        win_length (int): Window size of Fourier fransform (default=2048)
        hop_length (int): Hop size of Fourier fransform (default=512)
        window  (str): Window function (default='hann')
        pad_mode (str): Padding mode used in windowing (default='constant' -> fill with zeros)
    Returns:
        stft: STFT magnitude matrix

getTonic(x)
    Get the tonic of the audio signal

    Args:
        x (np.array): Audio signal vector (numpy array)
    Returns:
        im (float): Tonic of the audio signal

loadAudio(filePath, sampleRate=22050)
    Load audio sample

    Args:
        filePath (str): Path of the audio file
        sampleRate (int): Sampling rate
    Returns:
        x (np.ndarray): Audio sample values sampled at sampleRate
        sr (int): Sampling rate

logCompression(x)
    Log compression the input value

    Args:
        x : Magnitude/ numpy array / numpy matrix
    Returns:
        im : Log compression of input
```